

Name:

Teacher:



Fort Street High School

2011 Assessment task 3

Mathematics

Time allowed: 90 minutes

Outcomes Assessed	Questions	Marks
Demonstrates the ability to manipulate and simplify numeric and algebraic expressions.	1	
Solves problems involving inequalities, indices and logs.	2	
Synthesises mathematical understanding and applies appropriate techniques to solve problems involving linear functions and parabolas.	3,4	
Manipulates algebraic expressions to solve problems with the locus and the parabola.	5,6	

Question	1	2	3	4	5	6	Total (60)	%
Marks								

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question.
Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- Each Question is to be started in a new booklet.

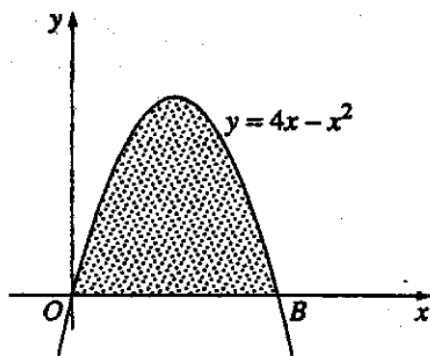
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- | | | |
|----|--|----------|
| a) | Evaluate $-7 - -5 $ | 1 |
| b) | Factorise $x^3 - 27$ | 1 |
| c) | Find the values of a and b if $\frac{5}{1+\sqrt{5}} = a + b\sqrt{5}$ | 2 |
| d) | Solve $ 2 - 3x < 5$ | 2 |
| e) | Simplify without negative indices $\frac{1+x^{-1}}{x-x^{-1}}$ | 2 |
| f) | Simplify $\sqrt{72} - \sqrt{50}$ | 2 |

a) Evaluate $\log_3 11$ correct to 2 significant figures. 1

b) Simplify $\frac{3\log_2 8}{\log_3 9}$ 2

c) Solve the equation $2(\log_2 x)^2 - 7\log_2 x + 3 = 0$ 3

d) The diagram shows the graph of the function $y = 4x - x^2$

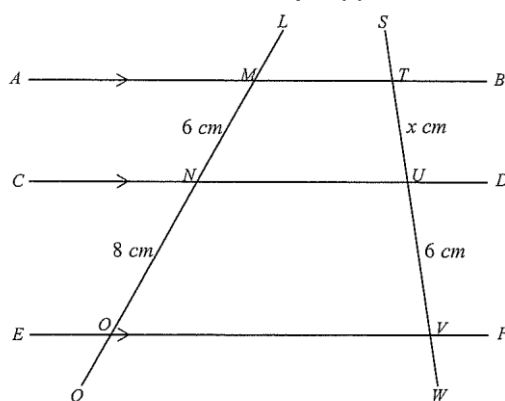


i) Find the range of the function. 2

ii) Write down a pair of inequalities that specify the shaded region. 2

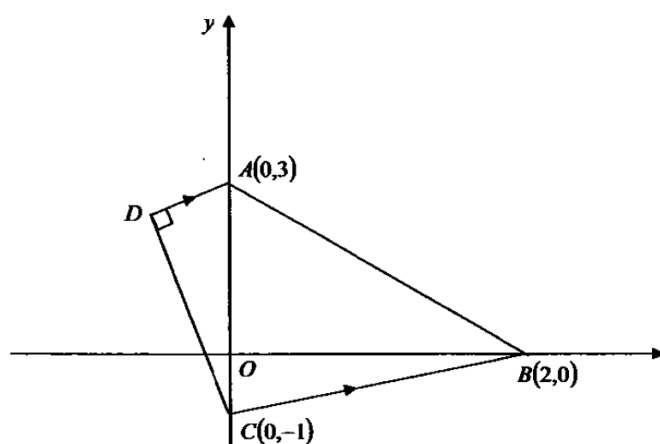
- a) Find the value of x giving a reason for your answer

3



- b) i) Sketch the parabola $x^2 = 12y$ clearly showing the focus and directrix. 2
- ii) Explain why an end point of the latus rectum is given by $(6, 3)$. 1
- iii) A line, l , through the focus of the parabola makes an angle of inclination of 45° with the x -axis. Find the equation of the line, l , in general form. 2
- iv) Hence find the shortest distance from $(6, 3)$ to the line l . 2
Give your answer in exact form.

- a) In the diagram, points A , B and C have coordinates $(0,3)$, $(2,0)$ and $(0,-1)$ respectively.
 $AD \parallel BC$ and $AD \perp CD$.



NOT TO SCALE

Copy this diagram into your answer booklet.

- | | | |
|-------|--|----------|
| (i) | Show that the gradient of the line BC is equal to $\frac{1}{2}$. | 1 |
| (ii) | Show that the equation of the line AD is $x - 2y + 6 = 0$. | 2 |
| (iii) | Find the equation of line CD . Give your answer in general form. | 2 |
| (iv) | By solving simultaneously the equations from (ii) and (iii), find the coordinates of point D . | 2 |
| (v) | Find the area of the quadrilateral $ABCD$. | 3 |

a) Let α and β be the roots of the equation $x^2 - 5x + 2 = 0$.

Find the values of

- | | | |
|------|---------------------------|---|
| i) | $\alpha + \beta$ | 1 |
| ii) | $\alpha\beta$ | 1 |
| iii) | $(\alpha + 1)(\beta + 1)$ | 2 |

b) The quadratic equation $P(x)$ is given by $P(x) = x^2 - 2(k - 3)x + (k - 1)$.

- | | | |
|-----|--|---|
| i) | Find the values of k for which $P(x) = 0$ has equal roots. | 2 |
| ii) | Find the range of values of k for which the quadratic function is positive definite. | 2 |

c) Find the value of m if one root of the function $y = x^2 - 2mx + (m + 3)$ is three times the other. 2

-
- a) If $ax(x-b) + (x-c) \equiv 2x^2 - x - 1$, find the values of a , b and c . **2**
- b) A point $P(x,y)$ moves so that its distance from the point $A(6,0)$ is twice its distance from the point $B(0,6)$. **3**
- Find the equation of the locus and describe it geometrically.
- c) A parabola has the equation $x = 12 + 4y - y^2$.
- i) Find the coordinates of the vertex **2**
- ii) Find the focal length **1**
- iii) Find the equation of the directrix **1**
- iv) Hence sketch $x = 12 + 4y - y^2$ **1**

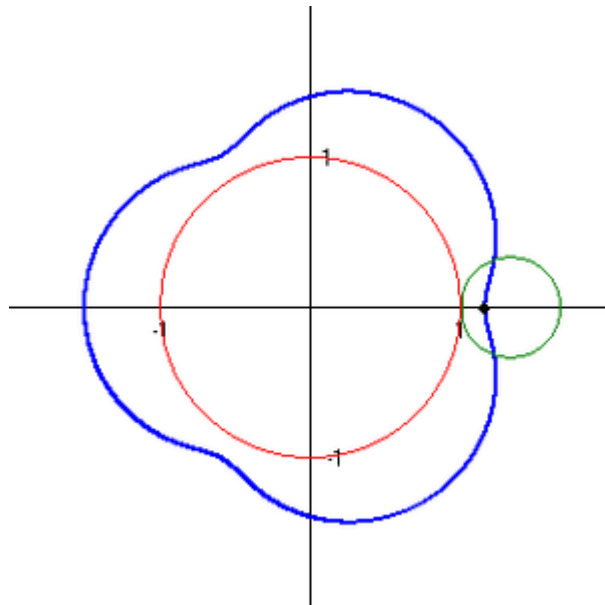


Fort Street High School

2011 Assessment task 3

Mathematics

Solutions



Question 1. (10 marks)

a) Evaluate $-7 - |-5|$

Solution

$$\begin{aligned} -7 - |-5| &= -7 - 5 \\ &= -12 \end{aligned}$$

Marking Guideline:

1 For correct response

b) Factorise $x^3 - 27$

Solution

$$x^3 - 27 = (x - 3)(x^2 + 3x + 9)$$

Marking Guideline:

1 For correct response

c) Find the values of a and b if $\frac{5}{1+\sqrt{5}} = a + b\sqrt{5}$

Solution

$$\begin{aligned} \frac{5}{1+\sqrt{5}} &= \frac{5}{1+\sqrt{5}} \times \frac{1-\sqrt{5}}{1-\sqrt{5}} \\ &= \frac{5-5\sqrt{5}}{1-5} \\ &= \frac{5-5\sqrt{5}}{-4} \\ &= -\frac{5}{4} + \frac{5}{4}\sqrt{5} \\ \Rightarrow a &= -\frac{5}{4}, b = \frac{5}{4} \end{aligned}$$

Marking Guideline:

2 For correct response

1 For finding $\frac{5-5\sqrt{5}}{1-5}$

d) Solve $|2 - 3x| < 5$

Solution

$$(2 - 3x) < 5 \quad \text{or} \quad (2 - 3x) > -5$$

$$-3x < 3$$

$$-3x > -7$$

$$x > -1$$

$$x < \frac{7}{3}$$

Marking Guideline:

2 For correct response

1 For one correct answer

e) Simplify without negative indices $\frac{1 + x^{-1}}{x - x^{-1}}$

Solution

$$\frac{1 + x^{-1}}{x - x^{-1}} = \frac{1 + \frac{1}{x}}{x - \frac{1}{x}}$$

$$= \frac{\frac{x}{x} + \frac{1}{x}}{x^2 - \frac{1}{x}}$$

$$= \frac{\frac{x+1}{x}}{\frac{x^2-1}{x}}$$

$$= \frac{x+1}{x} \times \frac{x}{x^2-1}$$

$$= \frac{x+1}{x^2-1}$$

$$= \frac{x+1}{(x+1)(x-1)}$$

$$= \frac{1}{x-1}$$

Marking Guideline:

2 For correct response

1 For finding $\frac{\frac{x+1}{x}}{\frac{x}{x^2-1}}$

f) Simplify $\sqrt{72} - \sqrt{50}$

Solution

$$\begin{aligned}\sqrt{72} - \sqrt{50} &= \sqrt{36} \times \sqrt{2} - \sqrt{25} \times \sqrt{2} \\ &= 6\sqrt{2} - 5\sqrt{2} \\ &= \sqrt{2}\end{aligned}$$

Marking Guideline:

2 For correct response

1 For finding $6\sqrt{2} - 5\sqrt{2}$

- a) Evaluate $\log_3 11$ correct to 2 significant figures.

Solution

$$\begin{aligned}\log_3 11 &= \frac{\log_{10} 11}{\log 3} \\ &= 2.2\end{aligned}$$

Correct to two significant figures.

Marking Guideline:

1 For correct response

- b) Simplify $\frac{3\log_2 8}{\log_3 9}$

Solution

$$\begin{aligned}\frac{3\log_2 8}{\log_3 9} &= \frac{3\log_2 2^3}{\log_3 3^2} \\ &= \frac{9\log_2 2}{2\log_3 3} \\ &= \frac{9}{2}\end{aligned}$$

Marking Guideline:

2 For correct response

1 For correct use of one log law

- c) Solve the equation $2(\log_2 x)^2 - 7\log_2 x + 3 = 0$

Solution

$$2(\log_2 x)^2 - 7\log_2 x + 3 = 0$$

let $m = \log_2 x$ then

$$2(m)^2 - 7m + 3 = 0$$

$$(2m-1)(m-3) = 0$$

$$2m-1=0$$

or

$$m-3=0$$

$$m = \frac{1}{2}$$

$$m = 3$$

but $m = \log_2 x$

so $\log_2 x = \frac{1}{2}$

or

$$\log_2 x = 3$$

$$x = 2^{\frac{1}{2}}$$

$$= \sqrt{2}$$

$$x = 2^3$$

$$= 8$$

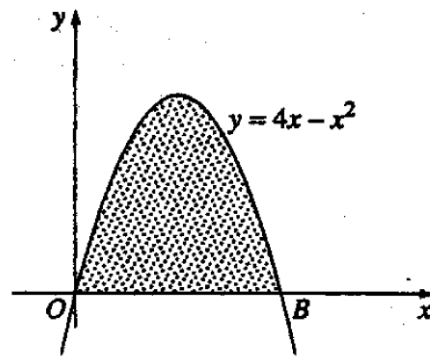
Marking Guideline:

3 For correct response or

2 For finding one correct answer or

1 For finding $(2m-1)(m-3)$

d) The diagram shows the graph of the function $y = 4x - x^2$



i) Find the range of the function.

Solution

The range will be all values of y less than (or equal to) the y ordinate of the vertex.

Finding the vertex:

$$x = \frac{-b}{2a}$$

$$= \frac{-(4)}{2(-1)}$$

$$= 2$$

When $x = 2$, $y = 4(2) - (2)^2$

$$= 4$$

Therefore the range is $y \leq 4$

Marking Guideline:

2 For correct response

1 For correct procedure with an error

ii) Write down a pair of inequalities that specify the shaded region.

Solution

The shaded region will be below the graph and above the line $y = 0$.

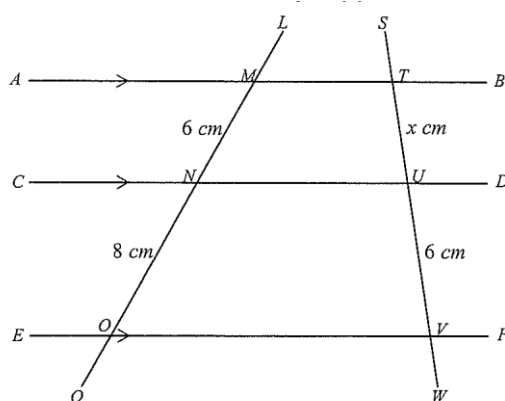
Therefore the inequality is $y \leq 4x - x^2$ and $y \geq 0$

Marking Guideline:

2 For correct response

1 For correct procedure with an error

- a) Find the value of x giving a reason for your answer



Solution

Intercepts of parallel lines cut by transversals are in equal ratio.

$$\frac{x}{6} = \frac{6}{8}$$

$$x = \frac{36}{8}$$

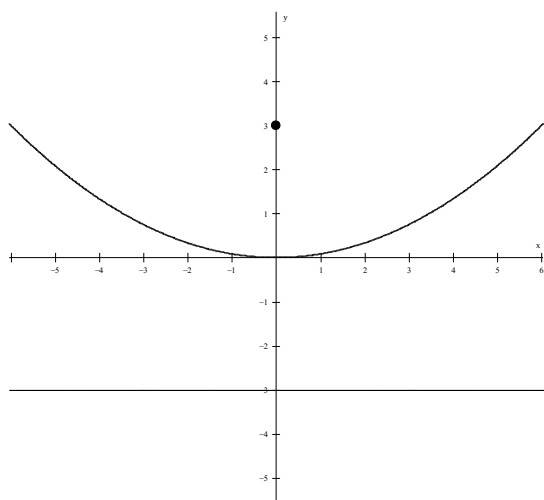
$$= 4.5$$

Marking Guideline:

- | | |
|---|---|
| 3 | For correct response or |
| 2 | For correct value with no reason or |
| 1 | For finding $\frac{x}{6} = \frac{6}{8}$ |

- b) i) Sketch the parabola $x^2 = 12y$ clearly showing the focus and directrix.

Solution



Marking Guideline:

- | | |
|---|----------------------|
| 2 | For correct response |
| 1 | For correct shape |

- ii) Explain why an end point of the latus rectum is given by $(6,3)$.

Solution

Method 1.

The length of the latus rectum is given by $4a$.

Since $a = 3$ the length would be $4 \times 3 = 12$ units.

Since the latus rectum is symmetrical about the axis, half the latus rectum would be 6 units.

This (half) length runs parallel to the x axis starting from the focus $(0,3)$. This places the end point at $(6,3)$

Method 2.

The equation of the latus rectum is $y = 3$ and is a focal chord of the parabola.

Substituting $y = 3$ into $x^2 = 12y$ to find the points of intersection

$$x^2 = 12(3)$$

$$x^2 = 36$$

$$x = \pm 6$$

Therefore one end point would be $(6,3)$

Marking Guideline:

1 For correct response

- iii) A line, l , through the focus of the parabola makes an angle of inclination of 45° with the x -axis.
Find the equation of the line, l , in general form.

Solution

The gradient of the line is found by $m = \tan \theta$

$$= \tan 45^\circ$$

$$= 1$$

The equation of the line is found by $y - y_1 = m(x - x_1)$

$$y - 3 = 1(x - 0)$$

$$x - y + 3 = 0$$

Marking Guideline:

2 For correct response

1 For finding the gradient

- iv) Hence find the shortest distance from $(6,3)$ to the line l .

Give your answer in exact form.

Solution

Shortest distance is perpendicular distance:

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

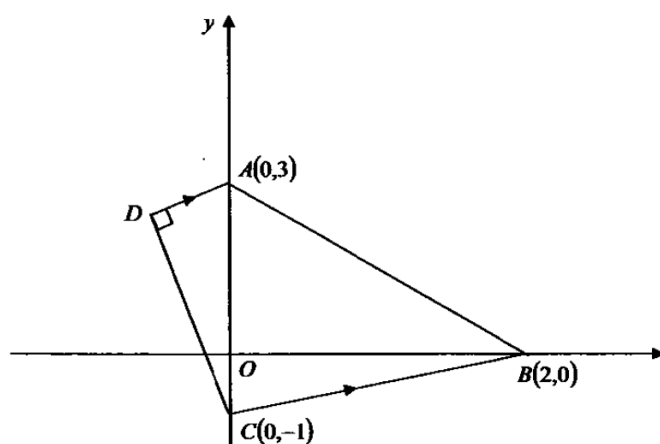
$$= \frac{|1(6) - 1(3) + 3|}{\sqrt{1^2 + (-1)^2}}$$

$$= \frac{6}{\sqrt{2}}$$

Marking Guideline:

- | | |
|---|---|
| 2 | For correct response |
| 1 | For correct procedure with arithmetic error |

- a) In the diagram, points A , B and C have coordinates $(0,3)$, $(2,0)$ and $(0,-1)$ respectively.
 $AD \parallel BC$ and $AD \perp CD$.



NOT TO SCALE

Copy this diagram into your answer booklet.

- (i) Show that the gradient of the line BC is equal to $\frac{1}{2}$.

Solution

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{0 - 3}{2 - 0} \\ &= -\frac{3}{2} \end{aligned}$$

Marking Guideline:

1 For correct response

- (ii) Show that the equation of the line AD is $x - 2y + 6 = 0$.

Solution

$$AD \text{ is parallel to } BC \Rightarrow m_{AD} = m_{BC} = \frac{1}{2}$$

$$y - y_1 = m(x - x_1)$$

$$y - 3 = \frac{1}{2}(x - 0)$$

$$x - 2y + 6 = 0$$

Marking Guideline:

2 For correct response

1 For not stating $m_{AD} = m_{BC} = \frac{1}{2}$

(iii) Find the equation of line CD . Give your answer in general form.

Solution

Since the line CD is perpendicular to AD the gradient of CD can be found using

$$m_1 m_2 = -1$$

$$\frac{1}{2} m_{CD} = -1$$

$$m_{CD} = -2$$

The equation of the line can now be found using the point gradient formula and point C

$$y - y_1 = m(x - x_1)$$

$$y - -1 = -2(x - 0)$$

$$2x + y + 1 = 0$$

Marking Guideline:

2 For correct response

1 For finding $m_{CD} = -2$

(iv) By solving simultaneously the equations from (ii) and (iii), find the coordinates of point D .

Solution

$$2x + y + 1 = 0 \quad ---1.$$

$$x - 2y + 6 = 0 \quad ---2.$$

1. $\times 2$

$$4x + 2y + 2 = 0 \quad ---1a.$$

$$x - 2y + 6 = 0 \quad ---2.$$

1a. $+ 2$

$$5x + 8 = 0$$

$$5x = -8$$

$$x = -\frac{8}{5}$$

$$\text{Sub. } x = -\frac{8}{5} \text{ into 2. } -\frac{8}{5} - 2y + 6 = 0$$

$$8 + 10y - 30 = 0$$

$$10y = 22$$

$$y = \frac{22}{10}$$

$$= 2.2$$

Therefore point D is $(-1.6, 2.2)$

Marking Guideline:

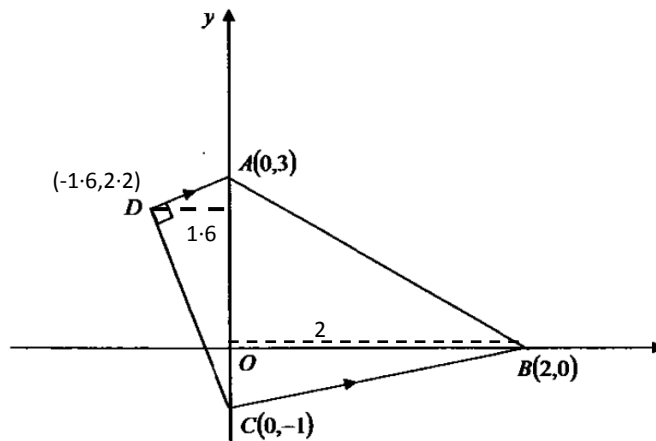
2 For correct response

1 For correct procedure with one error

(v) Find the area of the quadrilateral $ABCD$.

Solution

Break the quadrilateral into two triangles ACD & ABC



NOT TO SCALE

$$\begin{aligned}
 \text{Area of triangle } ABC &= \frac{1}{2}bh \\
 &= \frac{1}{2} \times OB \times AC \\
 &= \frac{1}{2} \times 2 \times 4 \\
 &= 4 \text{ } u^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of triangle } ACD &= \frac{1}{2}bh \\
 &= \frac{1}{2} \times AD \times AC \\
 &= \frac{1}{2} \times 1.6 \times 4 \\
 &= 3.2 \text{ } u^2
 \end{aligned}$$

$$\text{Total area of quadrilateral } ABCD = 7.2 \text{ } u^2$$

Marking Guideline:

- | | |
|---|--|
| 3 | For correct response or |
| 2 | For correct procedure with one arithmetic error or |
| 1 | For finding area ABC |

a) Let α and β be the roots of the equation $x^2 - 5x + 2 = 0$.

Find the values of

i) $\alpha + \beta$

Solution

$$\begin{aligned}\alpha + \beta &= \frac{-b}{a} \\ &= \frac{-(-5)}{1} \\ &= 5\end{aligned}$$

Marking Guideline:

1 For correct response

ii) $\alpha\beta$

Solution

$$\begin{aligned}\alpha\beta &= \frac{c}{a} \\ &= \frac{2}{1} \\ &= 2\end{aligned}$$

Marking Guideline:

1 For correct response

iii) $(\alpha + 1)(\beta + 1)$

Solution

$$\begin{aligned}(\alpha + 1)(\beta + 1) &= \alpha\beta + \alpha + \beta + 1 \\ &= 2 + 5 + 1 \\ &= 8\end{aligned}$$

Marking Guideline:

2 For correct response

1 For expanding $(\alpha + 1)(\beta + 1)$

b) The quadratic equation $P(x)$ is given by $P(x) = x^2 - 2(k - 3)x + (k - 1)$.

i) Find the values of k for which $P(x) = 0$ has equal roots.

Solution

Method 1. The quadratic has equal roots when the discriminant is 0.

Note: $\Delta = b^2 - 4ac$

$$b^2 - 4ac = 0$$

$$[-2(k-3)]^2 - 4.1.(k-1) = 0$$

$$4(k-3)^2 - 4k + 4 = 0$$

$$4(k^2 - 6k + 9) - 4k + 4 = 0$$

$$k^2 - 6k + 9 + 1 = 0$$

$$k^2 - 7k + 10 = 0$$

$$(k-5)(k-2) = 0$$

$$k = 5 \quad \text{or} \quad k = 2$$

Method 2. Let the roots be α and α . Then

$$\alpha + \alpha = \frac{-b}{a}$$

$$\alpha.\alpha = \frac{c}{a}$$

$$2\alpha = \frac{-[-2(k-3)]}{1}$$

$$\alpha^2 = \frac{k-1}{1}$$

$$\alpha = k-3$$

$$\alpha^2 = k-1$$

sub α into α^2

$$(k-3)^2 = k-1$$

$$k^2 - 6k + 9 = k-1$$

$$k^2 - 7k + 10 = 0$$

$$(k-5)(k-2) = 0$$

$$k = 5 \quad \text{or} \quad k = 2$$

Marking Guideline:

2 For correct response

1 For correct procedure with one arithmetic error

- ii) Find the range of values of k for which the quadratic function is positive definite.

Solution

The quadratic is positive definite if the discriminant is less than 0.

From part i) the discriminant simplifies to $k^2 - 7k + 10$

$$k^2 - 7k + 10 < 0$$

$$(k - 5)(k - 2) < 0$$

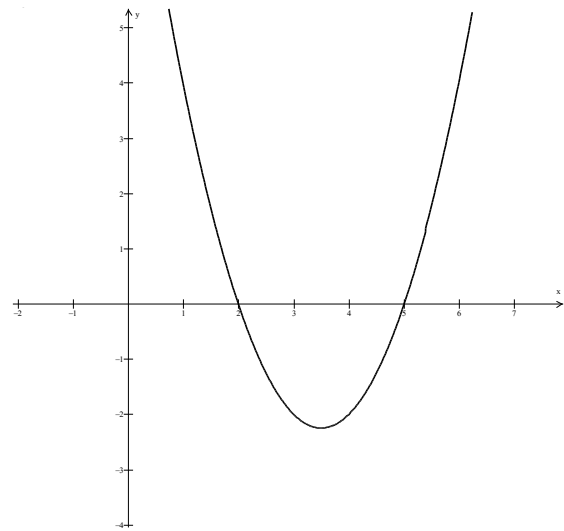
From the graph of $k^2 - 7k + 10$ this occurs when

$$2 < k < 5$$

Marking Guideline:

2 For correct response

1 For correct procedure with one error



Note: this is not the graph of $P(x)$
but $k^2 - 7k + 10$

- c) Find the value of m if one root of the function $y = x^2 - 2mx + (m + 3)$ is three times the other.

Solution

Let the roots be α and 3α then

$$\alpha + 3\alpha = \frac{-b}{a}$$

$$\alpha \cdot 3\alpha = \frac{c}{a}$$

$$4\alpha = \frac{-(-2m)}{1}$$

$$3\alpha^2 = \frac{m+3}{1}$$

$$\alpha = \frac{2m}{4}$$

$$\alpha^2 = \frac{m+3}{3}$$

$$= \frac{m}{2}$$

Substitute $\alpha = \frac{m}{2}$ into the equation for α^2

$$\left(\frac{m}{2}\right)^2 = \frac{m+3}{3}$$

$$\frac{m^2}{4} = \frac{m+3}{3}$$

$$3m^2 = 4m + 12$$

$$3m^2 - 4m - 12 = 0$$

$$m = \frac{-(-4) \pm \sqrt{16 - 4.3.(-12)}}{2.3}$$

$$= \frac{4 \pm \sqrt{160}}{6}$$

$$= \frac{4 \pm 4\sqrt{10}}{6}$$

$$= \frac{2 \pm 2\sqrt{10}}{3}$$

Marking Guideline:

2 For correct response

1 For correct procedure with one error

- a) If $ax(x-b) + (x-c) \equiv 2x^2 - x - 1$, find the values of a , b and c .

Solution

$$ax(x-b) + (x-c) \equiv 2x^2 - x - 1$$

$$ax^2 - abx + x - c \equiv 2x^2 - x - 1$$

$$ax^2 + (1-ab)x - c \equiv 2x^2 - x - 1$$

Comparing coefficients

$$c = 1, a = 2 \quad 1 - ab = -1$$

$$1 - (2)b = -1$$

$$-2b = -2$$

$$b = 1$$

Marking Guideline:

2 For correct response

1 For correct procedure with one error

- b) A point $P(x,y)$ moves so that its distance from the point $A(6,0)$ is twice its distance from the point $B(0,6)$.

Find the equation of the locus and describe it geometrically.

Solution

$$PA = 2PB$$

$$(PA)^2 = 4(PB)^2$$

$$(x-6)^2 + (y-0)^2 = 4[(x-0)^2 + (y-6)^2]$$

$$x^2 - 12x + 36 + y^2 = 4x^2 + 4y^2 - 48y + 144$$

$$3x^2 + 12x + 3y^2 - 48y + 108 = 0$$

$$x^2 + 4x + y^2 - 16y + 36 = 0$$

$$x^2 + 4x + y^2 - 16y = -36$$

$$x^2 + 4x + 4 + y^2 - 16y + 64 = -36 + 4 + 64$$

$$(x+2)^2 + (y-8)^2 = 32$$

Marking Guideline:

3 For correct response or

2 For correct locus without geometric description or

1 For demonstrating some understanding

This is the locus of a circle whose centre is $(-2,8)$ with a radius of $\sqrt{32}$ units.

c) A parabola has the equation $x = 12 + 4y - y^2$.

i) Find the coordinates of the vertex

Solution

The parabola has the form $(y - k)^2 = -4a(x - h)$

Rewriting

$$x = 12 + 4y - y^2$$

$$-x = y^2 - 4y - 12$$

$$-x + 12 + 4 = y^2 - 4y + 4$$

$$(y - 2)^2 = -x + 16$$

$$(y - 2)^2 = -1(x - 16)$$

This implies the vertex is at $(16, 2)$

Marking Guideline:

2 For correct response

1 For correct procedure with one error

ii) Find the focal length

Solution

$$4a = 1$$

$$a = \frac{1}{4}$$

Marking Guideline:

1 For correct response

iii) Find the equation of the directrix

Solution

Since the vertex is at $(16, 2)$ and in the form $(y - k)^2 = -4a(x - h)$

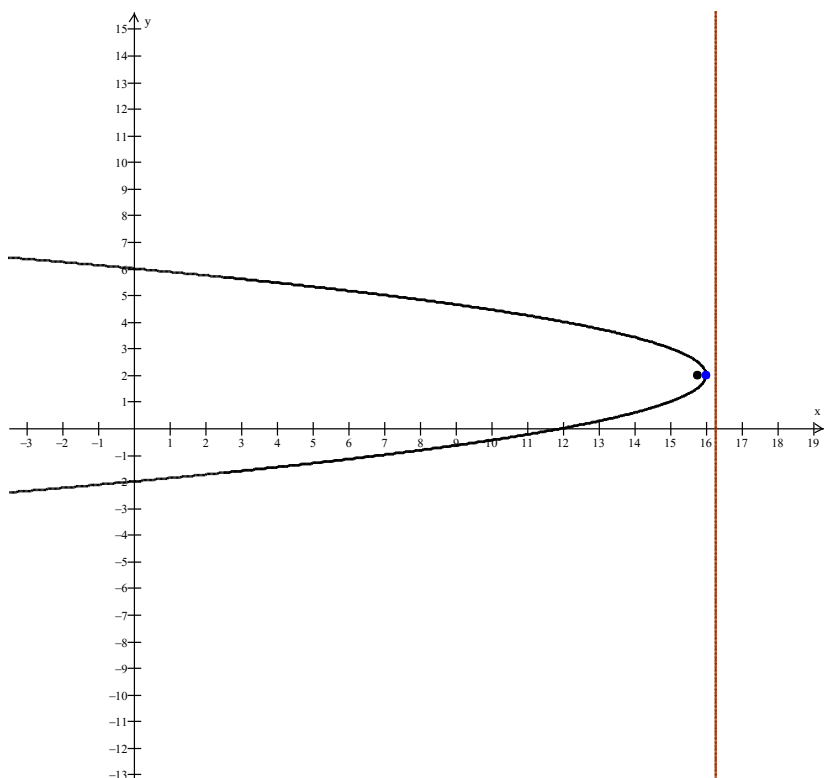
The equation of the directrix will be parallel to the y axis and $\frac{1}{4}$ units to the right of the vertex

Therefore the equation of the directrix is $x = 16\frac{1}{4}$

Marking Guideline:

1 For correct response

iv) Hence sketch $x = 12 + 4y - y^2$



Marking Guideline:

1 For correct response